Multi-asset benchmark — results (53/54 arms final)

Coinbase BTC / ETH / SOL, 7 days Jan 2026 — final_v2 protocol per coin

Snapshot 2026-07-13. 6 models × 3 training seeds × 3 rollout seeds per coin; seq/stride 1024, 12 epochs (equal step counts across models within each coin), val-split rate calibration with full-scale verification (SAHP uncalibrated by protocol, †), lossless streaming prediction, carried-state rollouts, equal-duration real-vs-sim comparison. **Status**: final for 53 of 54 arms. The single outstanding arm, eth/ss2p2-full-s1 (delayed by a faulty compute node, now retraining on an A100), is excluded from this report; all other numbers are final. All calibration exclusions are confirmed reproducible (see §4).

1. Run status

coin	complete	train fail (infra)	SF fail (calibration)	running (incl. reruns)
btc	15	0	3	0
eth	15	0	2	1
sol	15	0	3	0

All infrastructure train failures (NFS stale-handle cache race, "CUDA device busy") were rerun successfully. The remaining failures are *substantive*, reproducible calibration failures, all in the S2P2 family (§4). The one running arm is eth/ss2p2-full-s1 (healthy, on an A100).

2. Prediction (per genuine event; mean ± 95% CI over available seeds)

Bold = best per column: lower is better except ACC (higher) and mean_u (closest to 1).

BTC

model	overall	timeNLL	markNLL	KS	mean_u	tMAE	ACC	PPL
nhp	**-1.067±0.014**	**-3.215±0.004**	**2.148±0.016**	**0.227±0.006**	1.07±0.00	**0.03±0.00**	**0.449±0.003**	**8.57±0.13**
lstm	-0.437±0.400	-2.710±0.343	2.273±0.063	0.305±0.034	1.44±0.54	0.03±0.00	0.411±0.015	9.71±0.62
sahp	-0.928±0.014	-3.189±0.019	2.261±0.006	0.256±0.023	0.90±0.06	0.03±0.00	0.412±0.001	9.59±0.06
pct-lstm	-0.598±0.079	-3.176±0.048	2.579±0.035	0.242±0.042	**1.01±0.11**	0.03±0.00	0.313±0.010	13.18±0.47
s2p2	-0.397±0.235	-2.578±0.245	2.181±0.019	0.330±0.037	2.36±0.21	1.47±2.69	0.439±0.003	8.86±0.17
ss2p2-full	-0.865±0.109	-3.018±0.116	2.152±0.008	0.265±0.001	1.85±0.13	0.03±0.00	0.447±0.003	8.61±0.07

ETH

model	overall	timeNLL	markNLL	KS	mean_u	tMAE	ACC	PPL
nhp	-0.812±0.033	-3.412±0.014	2.600±0.019	0.254±0.012	**1.03±0.05**	**0.03±0.00**	0.341±0.006	13.46±0.26
lstm	0.154±0.038	-2.606±0.029	2.760±0.009	0.336±0.005	1.66±0.01	0.03±0.00	0.288±0.003	15.80±0.14
sahp	-0.598±0.008	-3.321±0.011	2.723±0.005	0.278±0.057	0.95±0.21	0.04±0.01	0.297±0.004	15.23±0.08
pct-lstm	-0.342±0.103	-3.302±0.077	2.960±0.029	0.247±0.069	1.12±0.15	0.03±0.00	0.248±0.007	19.30±0.55
s2p2	-0.381±0.224	-3.011±0.197	2.629±0.027	0.309±0.015	2.23±0.26	10.16±31.00	0.331±0.008	13.87±0.38
ss2p2-full (n=2)	**-0.849±0.180**	**-3.429±0.160**	**2.581±0.020**	**0.212±0.001**	1.66±0.26	0.03±0.01	**0.344±0.009**	**13.21±0.26**

SOL

model	overall	timeNLL	markNLL	KS	mean_u	tMAE	ACC	PPL
nhp	0.528±0.014	-2.707±0.004	3.235±0.014	0.321±0.008	**1.15±0.06**	0.05±0.00	0.184±0.004	25.40±0.35
lstm	1.343±0.250	-2.067±0.218	3.410±0.040	0.420±0.025	1.56±0.45	0.06±0.00	0.141±0.005	30.28±1.22
sahp	0.668±0.024	-2.687±0.027	3.355±0.019	0.376±0.017	0.83±0.04	0.06±0.00	0.155±0.003	28.65±0.54
pct-lstm	0.895±0.028	-2.607±0.026	3.502±0.009	0.314±0.010	1.23±0.02	0.06±0.00	0.105±0.004	33.18±0.30
s2p2	0.671±0.494	-2.560±0.495	3.231±0.001	0.204±0.017	2.08±0.52	46.18±95.99	0.182±0.002	25.30±0.03
ss2p2-full	**0.274±0.047**	**-2.911±0.044**	**3.185±0.003**	**0.173±0.007**	1.57±0.05	**0.05±0.00**	**0.192±0.001**	**24.17±0.07**

3. Stylized facts (rel-err vs real; rollout seeds averaged per checkpoint; mean ± 95% CI over checkpoints)

Checkpoints whose calibration failed verification produce no SF files and are excluded with a note; † = SAHP reported uncalibrated (k=1) by protocol.

BTC

model	sim_rate	rate_re	Fano_re	clus_re	retACF_re	cal_k
nhp	37.92±2.81	0.086±0.080	0.863±0.069	**0.275±0.042**	**0.209±0.138**	0.800±0.090
lstm	38.51±3.24	0.103±0.093	0.993±0.000	0.495±0.404	0.537±0.447	0.472±0.060
sahp †	10.26±13.17	0.706±0.377	**0.763±0.200**	0.921±2.546	0.745±2.183	1.000±0.000
pct-lstm	37.13±1.90	**0.063±0.054**	0.786±0.119	0.360±0.288	0.333±0.321	0.728±0.092
s2p2 (n=1)	33.74	0.069	1.114	0.326	0.642	0.541
ss2p2-full (n=2)	43.29±63.90	0.240±1.830	0.788±0.992	0.462±0.618	0.384±1.139	0.608±0.167

Excluded (calibration failed at matched fidelity): s2p2-s1, s2p2-s2, ss2p2-full-s3

ETH

model	sim_rate	rate_re	Fano_re	clus_re	retACF_re	cal_k
nhp	47.98±7.02	0.373±0.201	0.865±0.110	0.518±0.227	**0.132±0.122**	0.805±0.000
lstm	48.78±0.89	0.396±0.025	0.994±0.000	0.765±0.294	0.497±0.510	0.545±0.000
sahp †	7.49±3.76	0.786±0.108	0.763±0.456	**0.435±0.927**	0.468±0.929	1.000±0.000
pct-lstm	49.56±9.14	0.418±0.262	**0.762±0.170**	0.546±0.504	0.367±0.472	0.762±0.174
s2p2 (n=1)	38.92	**0.267**	6.045	0.595	0.454	1.102
ss2p2-full (n=2)	48.96±25.83	0.401±0.739	0.783±0.221	0.827±0.452	0.663±0.040	0.644±0.796

Excluded (calibration failed at matched fidelity): s2p2-s1, s2p2-s2

SOL

model	sim_rate	rate_re	Fano_re	clus_re	retACF_re	cal_k
nhp	23.93±0.69	**0.192±0.034**	0.751±0.167	0.721±0.109	0.806±0.071	0.799±0.025
lstm	24.66±0.75	0.228±0.037	0.988±0.000	0.969±0.032	0.861±0.100	0.500±0.000
sahp †	6.40±3.02	0.681±0.150	0.925±0.078	**0.656±0.241**	0.859±0.063	1.000±0.000
pct-lstm	24.58±4.80	0.224±0.239	**0.460±0.335**	0.715±1.356	1.665±2.536	0.626±0.180
s2p2			*no calibrated checkpoints*			
ss2p2-full	26.08±3.37	0.298±0.168	0.598±0.599	0.683±0.624	**0.778±0.156**	0.591±0.080

Excluded (calibration failed at matched fidelity): s2p2-s1, s2p2-s2, s2p2-s3

4. Calibration findings and pending work

s2p2 near-criticality replicates and worsens with event rate. At Coinbase rates (24–48 ev/s), **7 of 9 s2p2 checkpoints failed val-side calibration** (btc s1,s2; eth s1,s2; sol s1,s2,s3) — vs 1 of 3 on Gemini ETH (∼4 ev/s). The signature is identical everywhere: a near-discontinuous closed-loop rate response in the rescaling constant k (e.g. btc-s1: k 0.5930→0.5938 flips the probe rate 36→48.5 ev/s) with full-scale verify misses of 30–143%. Every RESUME retry reproduced its failure with a *bit-identical* bisection bracket (probes are seeded deterministically), so these are properties of the checkpoints, not sampling noise. On SOL all three seeds failed: **s2p2 has no calibrated simulation arm on SOL** — a model-level exclusion, as with SAHP. The two surviving s2p2 arms are btc-s3 and eth-s3 (k=1.102, verified). **ss2p2-full lost exactly one checkpoint** (btc-s3, probe bisection non-convergence, also reproduced); the other eight calibrated and verified. **Pending**: one arm (eth/ss2p2-full-s1) retraining after two placements on a pathological node (13h tensor-cache build, then a 3.5h hang inside a 13.5GB cache load); it is now on an A100 with the cache loading in 70s, and the strict per-coin `REPORT_OK` runs follow its completion.

Reading the snapshot. Coinbase rates are roughly 6–13× Gemini ETH (val rates: btc 38.3, eth 47.9, sol 24.0 ev/s), so the fixed 1024-event context spans only ∼20–40s of market time here vs ∼4.5 min on Gemini — long-memory facts (clus/retACF) lean harder on carried state and TBPTT. Numbers are comparable *within* a coin; cross-coin comparisons should use relative errors only. CIs over fewer than 3 checkpoints are marked with their n; n=2 uses $t_{0.975}$=12.7 and is wide by construction.